

Rice University
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jdossgollin 
James Doss-Gollin

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RESEARCH INTERESTS

My research builds probabilistic hazard models that enable climate risk analysis and bottom-up decision support for complex infrastructure systems, with a focus on urban flooding. I integrate Earth science, artificial intelligence, and decision analysis to examine non-stationary hazards, assess their effects on built environments, and develop adaptive risk-management approaches.

APPOINTMENTS

Rice University	
Assistant Professor, Department of Civil & Environmental Engineering	2021–
Affiliated Faculty, Severe Storm Prediction, Education, and Evacuation from Disasters (SSPEED) Center	2021–
Affiliated Faculty, Ken Kennedy Institute	2023–
The Pennsylvania State University	
Postdoctoral Scholar, Earth & Environmental Systems Institute	2020

EDUCATION

Columbia University	
Ph.D. in Earth & Environmental Engineering	2020
M.S. in Earth & Environmental Engineering	2016
Yale University	
B.S. in Mechanical Engineering	2015

AWARDS AND HONORS

NSF CAREER Award , National Science Foundation	2026
Outstanding Reviewer Award , Earth's Future	2023
Nickolas and Liliana Themelis Fellowship , Fu Foundation School of Engineering and Applied Science, Columbia University	2018
Graduate Research Fellowship, Climate and Large-Scale Atmospheric Dynamics , National Science Foundation	2017
Presidential Distinguished Fellowship , Fu Foundation School of Engineering and Applied Science, Columbia University	2015
Distinction in Major , Department of Mechanical Engineering and Materials Science, Yale University	2015
Legacy Award , New Haven Promise	2015
Larry Coben '79 Fellowship , Yale University	2014
Vance-Carter Travel Award , Yale University	2013
Thomas C. Barry Travel Award , Yale University	2012

ADVISEE AWARDS

Dongwook Kim : Karen and John Huff Graduate Fellowship in Civil and Environmental Engineering.	2025
Yuchen Lu : H.W. Reeves Endowed Scholarship.	2022

PUBLICATIONS

† denotes Rice advisee publication. Google Scholar citations: 841, h-index: 14, i10-index: 17.

IN PREP. / UNDER REVIEW

- [2] Hancock, C. L., Dee, S. G., Haider, M. R., **Doss-Gollin, J.**, Lehner, F., Murphy, K., and Munoz, S. E. : —
Robust 21st Century Hydrological Trends in the Mississippi River Basin from CMIP6: West-Gets-Drier, East-Gets-Wetter.
- [1] Pollack, A., Benedict, J., Deb, M., **Doss-Gollin, J.**, Judi, D., Lehman, W., Lutz, N., Reesman, C., —
 Sarazen, E., Son, Y., Srikrishnan, V., Sun, N., and Keller, K. : *Unrefined National Building Inventories Can Mislead Risk Assessments and Decisions.* . DOI: [10.2139/ssrn.5575271](https://doi.org/10.2139/ssrn.5575271)

JOURNAL ARTICLES

- [27] Baer, J. A., Sebastian, A., Grimley, L. E., **Doss-Gollin, J.**, Wright, D. B., Hussain, M. A., and Webber, M. K. : "Neglecting Spatiotemporal Rainfall Variability Misrepresents Flood Hazard and Risk". *npj Natural Hazards*. DOI: [10.1038/s44304-026-00208-5](https://doi.org/10.1038/s44304-026-00208-5) 2026
- [26] Murphy, K., Dee, S., Hancock, C., Pitchon, E., **Doss-Gollin, J.**, Wallace, E., and Muñoz, S. : "The Changing Relationship between the Bermuda High, Great Plains Low-Level Jet, and Mississippi River Basin Hydroclimate from the Last Millennium to 2100". *Journal of Geophysical Research: Atmospheres*. DOI: [10.1029/2025JD044340](https://doi.org/10.1029/2025JD044340) 2026
- [25] O'Donnell, M., Dee, S., **Doss-Gollin, J.**, and Muñoz, S. E. : "Hydrologic Whiplash in the Mississippi River Basin: Mechanisms and Projections". *Geophysical Research Letters*. DOI: [10.1029/2026GL122807](https://doi.org/10.1029/2026GL122807) 2026
- [24] Pollack, A. B., Auermuller, L., Burleyson, C. D., Campbell, J., Condon, M., Cooper, C., Coronese, M., Dangendorf, S., **Doss-Gollin, J.**, Hegde, P., Helgeson, C., Kopp, R. E., Kwakkel, J., Lesk, C., Mankin, J., Nicholas, R. E., Rice, J., Roth, S., Srikrishnan, V., Scheeler, M., Tuana, N., Vernon, C., Zhao, M., and Keller, K. : "Unlocking the Benefits of Transparent and Reusable Science for Climate Risk Management". *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2422157123](https://doi.org/10.1073/pnas.2422157123) 2026
- [23] Haider, M. R., Dee, S., **Doss-Gollin, J.**, Dunne, K., and Muñoz, S. : "Impact of 21st Century Climate Change on Mississippi River Basin Discharge in CESM2 Large Ensemble Projections". *Global and Planetary Change*. DOI: [10.1016/j.gloplacha.2025.104742](https://doi.org/10.1016/j.gloplacha.2025.104742) 2025
- [22] Liu, C., Kowal, D. R., **Doss-Gollin, J.**, and Vannucci, M. : "Bayesian Functional Graphical Models with Change-Point Detection". *Computational Statistics and Data Analysis*. DOI: [10.1016/j.csda.2024.108122](https://doi.org/10.1016/j.csda.2024.108122) 2025
- [21] Liu, Y., **Doss-Gollin, J.**, Dai, Q., Veeraraghavan, A., and Balakrishnan, G. : "Downscaling Extreme Precipitation with Wasserstein Regularized Diffusion". *IEEE Transactions on Geoscience and Remote Sensing*. DOI: [10.1109/TGRS.2025.3611872](https://doi.org/10.1109/TGRS.2025.3611872) 2025
- [20] **Lu, Y.†**, Seiyon Lee, B., and **Doss-Gollin, J.** : "Bayesian Spatiotemporal Nonstationary Model Quantifies Robust Increases in Daily Extreme Rainfall across the Western Gulf Coast". *Environmental Research: Climate*. DOI: [10.1088/2752-5295/adf56e](https://doi.org/10.1088/2752-5295/adf56e) 2025
- [19] O'Donnell, M., Murphy, K., **Doss-Gollin, J.**, Dee, S., and Munoz, S. : "Evaluation of Hydroclimatic Biases in the Community Earth System Model (CESM1) within the Mississippi River Basin". *Hydrology and Earth System Sciences*. DOI: [10.5194/hess-29-4637-2025](https://doi.org/10.5194/hess-29-4637-2025) 2025
- [18] Pollack, A., **Doss-Gollin, J.**, Srikrishnan, V., and Keller, K. : "UNSAFE: An UNcertain Structure and Fragility Ensemble Framework for Property-Level Flood Risk Estimation". *Journal of Open Source Software*. DOI: [10.21105/joss.07527](https://doi.org/10.21105/joss.07527) 2025
- [17] Kazadi, A., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : "FloodGNN-GRU: A Spatio-Temporal Graph Neural Network for Flood Prediction". *Environmental Data Science*. DOI: [10.1017/eds.2024.19](https://doi.org/10.1017/eds.2024.19) 2024
- [16] Murphy, K., Dee, S., **Doss-Gollin, J.**, Dunne, K. B. J., O'Donnell, M., and Muñoz, S. : "Competing Influences of Land Use and Greenhouse Gas Emissions on Mississippi River Basin Hydroclimate Simulated over the Last Millennium". *Paleoceanography and Paleoclimatology*. DOI: [10.1029/2024PA004902](https://doi.org/10.1029/2024PA004902) 2024
- [15] Singh, D., Bekris, Y. S., Rogers, C. D. W., **Doss-Gollin, J.**, Coffel, E. D., and Kalashnikov, D. A. : "Enhanced Solar and Wind Potential during Widespread Temperature Extremes across the U.S. Interconnected Energy Grids". *Environmental Research Letters*. DOI: [10.1088/1748-9326/ad2e72](https://doi.org/10.1088/1748-9326/ad2e72) 2024

- [14] Amonkar, Y., **Doss-Gollin, J.**, Farnham, D. J., Modi, V., and Lall, U. : “Differential Effects of Climate Change on Average and Peak Demand for Heating and Cooling across the Contiguous USA”. *Communications Earth & Environment*. DOI: [10.1038/s43247-023-01048-1](https://doi.org/10.1038/s43247-023-01048-1) 2023
- [13] Amonkar, Y., **Doss-Gollin, J.**, and Lall, U. : “Compound Climate Risk: Diagnosing Clustered Regional Flooding at Inter-Annual and Longer Time Scales”. *Hydrology*. DOI: [10.3390/hydrology10030067](https://doi.org/10.3390/hydrology10030067) 2023
- [12] **Doss-Gollin, J.**, Amonkar, Y., **Schmeltzer, K.**[†], and Cohan, D. : “Improving the Representation of Climate Risks in Long-Term Electricity Systems Planning: A Critical Review”. *Current Sustainable/Renewable Energy Reports*. DOI: [10.1007/s40518-023-00224-3](https://doi.org/10.1007/s40518-023-00224-3) 2023
- [11] **Doss-Gollin, J.** and Keller, K. : “A Subjective Bayesian Framework for Synthesizing Deep Uncertainties in Climate Risk Management”. *Earth’s Future*. DOI: [10.1029/2022EF003044](https://doi.org/10.1029/2022EF003044) 2023
- [10] Garcia, M., Juan, A., **Doss-Gollin, J.**, and Bedient, P. : “Leveraging Mesh Modularization to Lower the Computational Cost of Localized Updates to Regional 2D Hydrodynamic Model Outputs”. *Engineering Applications of Computational Fluid Mechanics*. DOI: [10.1080/19942060.2023.2225584](https://doi.org/10.1080/19942060.2023.2225584) 2023
- [9] Wutich, A., Thomson, P., Jepson, W., Stoler, J., Cooperman, A. D., **Doss-Gollin, J.**, Jantrania, A., Mayer, A., Nelson-Núñez, J., Walker, W. S., and Westerhoff, P. : “MAD Water: Integrating Modular, Adaptive, and Decentralized Approaches for Water Security in the Climate Change Era”. *WIREs Water*. DOI: [10.1002/wat2.1680](https://doi.org/10.1002/wat2.1680) 2023
- [8] Zhou, X., Duenas-Osorio, L., **Doss-Gollin, J.**, Liu, L., Stadler, L., and Li, Q. : “Mesoscale Modeling of Distributed Water Systems Enables Policy Search”. *Water Resources Research*. DOI: [10.1029/2022WR033758](https://doi.org/10.1029/2022WR033758) 2023
- [7] **Doss-Gollin, J.**, Farnham, D. J., Lall, U., and Modi, V. : “How Unprecedented Was the February 2021 Texas Cold Snap?”. *Environmental Research Letters*. DOI: [10.1088/1748-9326/ac0278](https://doi.org/10.1088/1748-9326/ac0278) 2021
- [6] **Doss-Gollin, J.**, Farnham, D. J., Ho, M., and Lall, U. : “Adaptation over Fatalism: Leveraging High-Impact Climate Disasters to Boost Societal Resilience”. *Journal of Water Resources Planning and Management*. DOI: [10.1061/\(asce\)wr.1943-5452.0001190](https://doi.org/10.1061/(asce)wr.1943-5452.0001190) 2020
- [5] **Doss-Gollin, J.**, Farnham, D. J., Steinschneider, S., and Lall, U. : “Robust Adaptation to Multiscale Climate Variability”. *Earth’s Future*. DOI: [10.1029/2019ef001154](https://doi.org/10.1029/2019ef001154) 2019
- [4] Rözer, V., Kreibich, H., Schröter, K., Müller, M., Sairam, N., **Doss-Gollin, J.**, Lall, U., and Merz, B. : “Probabilistic Models Significantly Reduce Uncertainty in Hurricane Harvey Pluvial Flood Loss Estimates”. *Earth’s Future*. DOI: [10.1029/2018ef001074](https://doi.org/10.1029/2018ef001074) 2019
- [3] **Doss-Gollin, J.**, Muñoz, Á. G., Mason, S. J., and Pastén, M. : “Heavy Rainfall in Paraguay during the 2015/16 Austral Summer: Causes and Subseasonal-to-Seasonal Predictive Skill”. *Journal of Climate*. DOI: [10.1175/JCLI-D-17-0805.1](https://doi.org/10.1175/JCLI-D-17-0805.1) 2018
- [2] Farnham, D. J., **Doss-Gollin, J.**, and Lall, U. : “Regional Extreme Precipitation Events: Robust Inference from Credibly Simulated GCM Variables”. *Water Resources Research*. DOI: [10.1002/2017WR021318](https://doi.org/10.1002/2017WR021318) 2018
- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Analytic Modeling of Rainwater Harvesting in the Brazilian Semiarid Northeast”. *Journal of the American Water Resources Association*. DOI: [10.1111/1752-1688.12376](https://doi.org/10.1111/1752-1688.12376) 2015

PEER-REVIEWED CONFERENCE PAPERS

- [4] Kazadi, A., **Doss-Gollin, J.**, and Silva, A. : “Pluvial Flood Emulation with Hydraulics-Informed Message Passing”. *Forty-First International Conference on Machine Learning*. URL: <https://openreview.net/forum?id=kIHIA6Lr0B> 2024
- [3] Kazadi, A. N., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : “Flood Prediction with Graph Neural Networks”. *Climate Change AI*. URL: <https://www.climatechange.ai/papers/neurips2022/75> 2022
- [2] Araújo Júnior, L. M., de Souza Filho, F. d. A., da Silva Silveira, C., Aragão Dias, T., and **Doss-Gollin, J.** : “Análise Dos Eventos de Seca No Nordeste Setentrional Brasileiro Com Base No Índice de Precipitação Normalizada”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4610.7685](https://doi.org/10.13140/rg.2.1.4610.7685) 2014

- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Considerações Sobre a Sustentabilidade Hídrica de Cisternas Para Captação de Chuva No Semiárido Brasileiro”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4086.4807](https://doi.org/10.13140/rg.2.1.4086.4807) 2014

SPONSORED RESEARCH

Amounts reflect Rice portion for collaborative grants and subawards; total amount for direct awards.

PI, National Science Foundation – Infrastructure Systems and People. CAREER: <i>Drain or Retain? Strategic Assessment of Integrated Flood Infrastructure Portfolios with Physics-Informed ML</i> . No. 2543303. (\$599,581)	2026–2031
Co-PI, National Science Foundation – Confronting Hazards, Impacts and Risks for a Resilient Planet (CHIRRP). <i>RAISE: Flood Resilience in Rural Texas Communities</i> PI: Avantika Gori, with Andrew Juan and Keri Stephens. No. 2440167. (\$999,986)	2025–2028
PI, Consortium for Enhancing Resilience and Catastrophe Modeling (CERCAT). <i>A Nonstationary Joint Probability Method for Tropical Cyclone Hazard Assessment</i> with Avantika Gori. No. P12. (\$75,000)	2025–2026
PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)	2025–2026
Co-PI, NVIDIA. <i>Computing Infrastructure for AI-enhanced Climate Risk and Resilience at Rice</i> PI: Arlei Lopes da Silva, with Avantika Gori. in-kind	2025–2025
PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)	2024–2025
Co-PI, National Science Foundation – Industry-University Cooperative Research Centers (IUCRC). <i>IUCRC Planning Grant Rice University: Center for Climate, Equity and Resilience in Catmodeling (CER-Cat)</i> PI: Jamie Padgett, with Katherine Ensor, Avantika Gori, Leonardo Dueñas Osorio, Brian Davison, David Casagrande, Maryam Rahnemoonfar, Paolo Bocchini, and Yi-Chen Yang. No. 2413382. (\$20,000)	2024–2025
Co-PI, Rice University Sustainability Institute. <i>Workshop on Nature-Based Solutions for Resilient Coastal Cities</i> PI: Philip Bedient, with Avantika Gori. (\$7,500)	2024–2025
PI, Texas Water Development Board. <i>Developing Future Rainfall Frequency Grids for the State of Texas</i> with John Nielsen-Gammon. (Full grant \$192,828. Subaward \$77,750)	2022–2026
Co-PI, National Science Foundation – Climate and Large-Scale Dynamics (CLD). <i>Collaborative Research: Evaluating the Past and Future of Mississippi River Hydroclimatology to Constrain Risk via Integrated Climate Modeling, Observations, and Reconstructions</i> PI: Sylvia Dee, with Samuel Muñoz. No. 2147781. (\$472,024)	2022–2025
PI, National Science Foundation – Strengthening America’s Infrastructure (SAI). <i>Collaborative Research: EAGER: Participatory Design for Water Quality Monitoring of Highly Decentralized Water Infrastructure Systems</i> with Alicia Cooperman, Alex Mayer, Shane Walker, and Manuela Muñoz Fuerte. No. 2120829. (\$85,046)	2022–2025
Co-PI, 100,000 Strong in the Americas Innovation Fund. <i>IFCE-Rice-SENAI Program on Artificial Intelligence for Urban Sustainability and Resilience to Natural Disasters in the Americas</i> PI: Arlei Lopes da Silva. (\$50,000)	2022–2023
Co-PI, Energy Foundation. <i>Synthesis of Texas Electricity Research from Rice University</i> PI: Daniel Cohan. (\$24,928)	2022–2023
PI, Rice University – Sustainable Futures Fund (SFF). <i>Leveraging Earth System Observations at Multiple Scales to Improve Stormwater Management in Houston</i> with Sylvia Dee. (\$50,000)	2022–2023

PRESENTATIONS

INVITED TALKS

“Statistical Modeling for Nonstationary Hydroclimate Hazards: Challenges and Opportunities”. Statistics Colloquium, Rice University , Houston, TX.	2025
Panelist. Exploring Water Resilience to Emerging Challenges, Society for Risk Analysis , Remote Presentation.	2025
“Advancing Urban Flood Risk Management through Physics-informed, Data-Driven Hazard Assessment”. Earth, Marine, and Environmental Science Seminar, University of North Carolina , Chapel Hill, NC.	2024

"Quantifying and Characterizing Uncertain Climate Hazards to Enable Adaptive Resilience". Atmospheric Sciences Seminar, Texas A&M University , College Station, TX.	2022
"Revisiting Our Design Criteria: What Hazards Should We Design for in a Changing Climate?". Hydrologic Sciences and Water Resources Engineering Seminar, University of Colorado Boulder , Remote Presentation.	2022
"Adapting Engineering Design Criteria to a Changing Climate: Insights from House Elevation". Technical Webinar, ASCE Central New Jersey Branch , Remote Presentation.	2022
Panelist. Extreme Weather: How To Report on a World That's Warmer, Colder, Wetter, Drier and Weirder, 31st Annual Conference of the Society of Environmental Journalists , Houston, TX.	2022
"Predictable and Preventable: Lessons in Climate Risk Management from the Texas Coast (and Beyond)". Climate Risk and Financial Institutions Seminar (Prof. Madison Condon), Boston University School of Law , Remote Presentation.	2022
"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Black-outs". Outreach Event: Science is for Everyone, American Meteorological Society , Remote Presentation.	2022
"Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Black-outs". Compound Events Working Group, Risk KAN (Knowledge Action Networks) , Remote Presentation.	2021
Panelist. Tail events: Prediction, Planning, and Performance, Harvard Electricity Policy Group , Remote Presentation.	2021
"Towards Adaptive Resilience: Managing Flood Risks in a Changing World". Technical Webinar, ASCE Central New Jersey Branch , Remote Presentation.	2021
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Center for Climate Risk Management CLIMA Seminar, the Pennsylvania State University , State College, PA.	2020
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Department of Civil and Environmental Engineering Seminar, Rice University , Houston, TX.	2020
"Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Complex Systems Simulation and Optimization Group, National Renewable Energy Laboratory , Golden, CO.	2020
"Drivers of Extreme Rainfall: Atmospheric Circulation Patterns and Regional Intense Rainfall in the Ohio River". European Flood Awareness System Group, European Centre for Medium Range Weather Forecasting , Reading, England.	2016
"Understanding the Physical Drivers of Extreme Rainfall for Flood Prediction". Oxford Water Network, Oxford University , Oxford, England.	2016
FIRST-AUTHOR CONFERENCE AND WORKSHOP PRESENTATIONS	
"Industry-Academia Collaborations on Seasonal and Long-Term TC Hazard Assessment (CERCat)". Catastrophe Risk Management Conference (CATRISK26) , <i>Reinsurance Association of America</i> , Orlando, FL [Panel].	2026
"H24A-07: TxRain: A Bayesian framework for integrating historical observations and model projections to develop nonstationary IDF curves". American Geophysical Union Fall Meeting 2025 , New Orleans, LA [Talk].	2025
"Robust Trends in Extreme Rainfall Probabilities in Texas". 2025 Texas Climate Conference , <i>Rice University and Texas A&M University</i> , Houston, TX [Talk].	2025
"Use-Inspired Tools for Climate Hazard Assessment". Nature-Based Solutions for a Resilient Gulf Coast Workshop , <i>Rice University</i> , Houston, TX [Talk].	2025
"Advancing Urban Flood Hazard Characterization through Machine Learning: Challenges and Opportunities". American Geophysical Union Fall Meeting 2024 , Washington, DC [Talk].	2024
"Assessing and Managing Climate Risks to Electricity Systems in an Era of Climate Change and Energy Transition". American Geophysical Union Fall Meeting 2024 , Washington, DC [Talk].	2024
"Leveraging Machine Learning to Advance Urban Flood Hazard Assessment: Challenges and Opportunities". Data-driven and physics-based machine learning methods for forecasting and knowledge discovery of surface hydrology , <i>Conference on Computational Methods in Water Resources</i> , Tucson, AZ [Talk].	2024

- "Incorporating Temperature Projections into Energy Systems Planning". **Extreme Heat Workshop**, Columbia University, New York, NY [Talk]. 2024
- "NH14B-07: Linking Robust Trends in Observations and Models to Develop Nonstationary Rainfall Frequency Grids for the State of Texas". **American Geophysical Union Fall Meeting 2023**, San Francisco, CA [Talk]. 2023
- "A Bayesian Spatial Hierarchical Framework for Process-Informed Nonstationary Analysis of Precipitation Frequencies". **13th International Workshop on Statistical Hydrology**, International Association of Hydrological Sciences, Boston, MA [Talk]. 2023
- "H35F-07: Near-term Predictability Lowers Long-Term Adaptation Costs". **American Geophysical Union Fall Meeting 2022**, Chicago, IL [Talk]. 2022
- "Unprecedented Impacts Don't Require Unprecedented Weather". **Post-Harvey Climate & Flood Impacts on the Built Environment**, Severe Storm Prediction, Education, & Evacuation from Disasters Center, Houston, TX. 2022
- "H25U-1265: Operationalizing Bayesian Model Checking for Robust Decision Making: Insights from House Elevation". **American Geophysical Union Fall Meeting 2021**, New Orleans, LA [Poster]. 2021
- "A14H-03: How Unprecedented Was the February 2021 Texas Cold Snap?". **American Geophysical Union Fall Meeting 2021**, New Orleans, LA [Talk]. 2021
- "Valuing Flexibility and Soft Instruments for Sequential Decision Problems". **2020 Annual Meeting**, Society for Decision Making under Deep Uncertainty, [Remote Presentation]. 2020
- "H11G-07: Towards Adaptive Resilience: Managing Uncertainties and Exploiting Predictability across Timescales". **American Geophysical Union Fall Meeting 2019**, San Francisco, CA [Talk]. 2019
- "Adaptive Resilience through Real Options and Deep Reinforcement Learning". **Doctoral Consortium on Computational Sustainability**, Carnegie Mellon University, Pittsburgh, PA [Talk]. 2019
- "Evaluating Staged Investments in Critical Infrastructure for Climate Adaptation". **2019 Interdisciplinary Ph.D. Workshop in Sustainable Development**, Columbia University, New York, NY [Talk]. 2019
- "H52F-05: Robust Adaptation to Cyclical Climate Risk". **American Geophysical Union Fall Meeting 2018**, Washington, DC [Talk]. 2018
- "Robust Adaptation to Multi-Scale Climate Variability". **The Nexus of Climate Data, Insurance, and Adaptive Capacity**, Asheville, NC [Poster]. 2018
- "A31H-0135: Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". **American Geophysical Union Fall Meeting 2017**, New Orleans, LA [Talk]. 2017
- "H22B-02: Designing and Operating Infrastructure for Nonstationary Flood Risk Management". **American Geophysical Union Fall Meeting 2017**, New Orleans, LA [Talk]. 2017
- "Extreme Rainfall in Paraguay during the 2015-16 Austral Summer: Causes and Predictive Skill". **North East Graduate Student Water Symposium**, University of Massachusetts Amherst, Amherst, MA [Talk]. 2017
- "Regional Intense Precipitation: Inferences From GCM Atmospheric Circulation Fields". **Modeling Research in the Cloud**, National Center for Atmospheric Research, Boulder, CO [Poster]. 2017
- "Statistical-Dynamical Analysis of Climate Projections for Flood Infrastructure Design". **Interdisciplinary Ph.D. Workshop in Sustainable Development 2017**, Columbia University, New York, NY [Talk]. 2017
- "Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". **Workshop on Subseasonal to Seasonal Predictability of Extreme Weather and Climate**, Columbia University, New York, NY [Poster]. 2016

ADVISEE CONFERENCE AND WORKSHOP PRESENTATIONS

- Yuchen Lu[†]**: "NH13D-0410: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". **American Geophysical Union Fall Meeting 2025**, New Orleans, LA [Poster]. 2025
- Yuchen Lu[†]**: "TxRAIN-Observational: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". **American Geophysical Union Fall Meeting 2024**, Washington, DC [Talk]. 2024
- Yuchen Lu[†]**: "Nonstationary Extreme Precipitation Probabilities in Texas". **Fall Meeting**, Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat), Bethlehem, PA [Poster]. 2024

Yuchen Lu[†] : “Spatially Varying and Duration Dependent Covariate Model: A Hierarchical Bayesian Framework for Multi-duration Extreme Precipitation Frequency Analysis in Texas”. World Environmental & Water Resources Congress 2024 , Environmental & Water Resources Institute (EWRI) - ASCE, Milwaukee, WI [Talk].	2024
Yuchen Lu[†] : “H21T-1602: Spatially Varying Covariate Model: A Hierarchical Bayesian Framework for Precipitation Frequency Analysis in the Gulf Coast”. American Geophysical Union Fall Meeting 2023 , San Francisco, CA [Talk].	2023
Yuchen Lu[†] : “H42E-1333: Nonstationary GEV with Hierarchical Spatial Pooling: A Spatiotemporal Bayesian Framework for Nonstationary Extreme Precipitation Frequency Analysis in the Gulf Coast”. American Geophysical Union Fall Meeting 2022 , Chicago, IL [Talk].	2022

TEACHING AND ADVISING

COURSES TAUGHT

Rice CEVE 543: <i>Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes</i> .	Fall 2026
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2026
Rice CEVE 543: <i>Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes</i> .	Fall 2025
Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .	Fall 2024
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2024
Rice CEVE 543: <i>Data Science Methods for Climate Hazard Assessment</i> .	Fall 2023
Rice CEVE 421/521: <i>Climate Risk Management</i> .	Spring 2023
Rice CEVE 543: <i>Environmental Data Science</i> .	Spring 2022
Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .	Fall 2021

CURRENT GRADUATE STUDENTS

True Furrh: Ph.D. in Civil and Environmental Engineering, Rice University. Committee Member .	2021–
Dongwook Kim: Ph.D. in Civil and Environmental Engineering, Rice University. Primary Advisor .	2024–
Kelsey Murphy: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Committee Member .	2021–
Jiayue Yin: Ph.D. in Earth, Environmental, and Planetary Sciences, Rice University. Committee Member .	2023–

PAST GRADUATE STUDENTS

Primary Advisor: Yuchen Lu, Ph.D. 2026; Katlyn Schmeltzer, MCEE 2025.

Committee Member: Valeriia Sobolevskaia, Ph.D. 2026; Karan Jakhar, Ph.D. 2025; Kyle Ostlind, M.S. 2025; John A. Baer, M.S. 2024; Kendall Capshaw, Ph.D. 2024; Xinyue Luo, Ph.D. 2024; Anibal Tafur Gutierrez, Ph.D. 2024; Matthew Garcia, Ph.D. 2023; Mia Peebles, M.S. 2023; Xiangnan Zhou, Ph.D. 2023; Raychel Bahnick, M.S. 2022; Alyssa Graham, M.S. 2022; Elizabeth Hoffmann, M.S. 2022; Chunshan Liu, Ph.D. 2022; Xiaoyu (Toby) Li, M.S. 2021.

UNDERGRADUATE RESEARCHERS

Year indicates graduation year.

Zain Rahman: <i>B.S. in Computer Science</i> , Rice University.	2027
Kyle Olcott: <i>B.S. in Civil and Environmental Engineering</i> , Rice University.	2025
Sophia Prieto: <i>B.S. in Statistics</i> , Rice University.	2023
John Cook: <i>B.S. in Civil and Environmental Engineering</i> , Rice University.	2022

TEAMS ADVISED

Optimal Policy for Decentralized Wastewater Systems (DWS) while Relaxing Certainty: Rice Computational Mathematics and Operations Research (CMOR) Senior Project.	2024-2025
Flood Sight Advancing Real-Time Flood Predictions for Situational Awareness: Rice Data to Knowledge (D2K) Lab.	Spring 2025
Predicting Streamflow for Hill Country Gauges Using a Transformer-based Model: Rice Data to Knowledge (D2K) Lab.	Fall 2025

SERVICE ACTIVITIES

DEPARTMENTAL SERVICE

Member , Graduate Studies Committee.	2024–
Member , Diversity, Equity, and Inclusion Committee.	2024
Member , Faculty Search Committee.	2022–2023
Member , Seminar Committee.	2022–2023

UNIVERSITY SERVICE

Member , Research Council. Ken Kennedy Institute.	2024–
Faculty Associate . Duncan College.	2023–
External Search Committee Member . Department of Earth, Environmental, and Planetary Sciences.	2022–2023

PROFESSIONAL SERVICE

Working Group Member , Digital Twins - Beyond Physical Systems. Digital Earths Lighthouse Activity.	2026–2030
Committee Member , Water and Society Newsletter Committee. American Geophysical Union.	2024–
Committee Member , Environmental and Water Resources Systems (EWRS) Committee. American Society of Civil Engineers (ASCE) Environmental & Water Resources Institute (EWRI).	2021–

PEER REVIEW

Journals: AGU Advances; Climate Risk Management; Climatic Change; Communications Earth and Environment; Earth's Future; Energy Technology; Engineering Applications of Computational Fluid Mechanics; Environmental Data Science; Environmental Research Letters; Environmental Research: Climate; Geophysical Research Letters; Hydrology and Earth System Sciences; IEEE Transactions on Geoscience and Remote Sensing; Joule; Journal of Applied Meteorology and Climatology; Journal of Catastrophe Risk and Resilience; Journal of Hydrology; Journal of Water Resources Planning and Management; Machine Learning: Earth; Natural Hazards and Earth System Sciences; npj Natural Hazards; Oxford Development Studies; Water Resources Research; Water Security; Weather, Climate, and Society.

Funding Agencies: Department of Energy (BER); Dutch Research Council (NWO); National Science Foundation.

Other: Electric Power Research Institute (EPRI); Texas Water Development Board (TWDB).

SESSIONS CONVENED

Co-Convenor . NH43C - <i>Advances in Urban Flood Risk Assessment and Adaptation IV Poster</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH43D - <i>Advances in Urban Flood Risk Assessment and Adaptation V Poster</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH41E - <i>Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice I Poster</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH43B - <i>Interdisciplinary Advances in Catastrophe Modeling and Disaster Resilience: Bridging Science, Policy, and Practice II Oral</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH31A - <i>Advances in Urban Flood Risk Assessment and Adaptation I Oral</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH32A - <i>Advances in Urban Flood Risk Assessment and Adaptation II Oral</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Convenor . NH33A - <i>Advances in Urban Flood Risk Assessment and Adaptation III Oral</i> . American Geophysical Union Fall Meeting , New Orleans, LA.	2025
Co-Organizer . <i>Nature-Based Solutions for a Resilient Gulf Coast</i> . Rice University , Houston, TX.	2025
Primary Convener . H31G - <i>Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security I Oral</i> . American Geophysical Union Fall Meeting , San Francisco, CA.	2023

Primary Convener. <i>H33P - Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security II Poster.</i> American Geophysical Union Fall Meeting , San Francisco, CA.	2023
Convener. <i>NH41C - Hybrid Modeling and Digital Twin Systems for Flood Hazard Prediction and Risk Assessment at Different Spatial Scales.</i> American Geophysical Union Fall Meeting , Washington, DC.	2023
Chair. <i>H44G - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making I Oral.</i> American Geophysical Union Fall Meeting , New Orleans, LA.	2021
Chair. <i>H45V - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making III Poster.</i> American Geophysical Union Fall Meeting , New Orleans, LA.	2021
Chair. <i>H45V - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making Poster Summary Session.</i> American Geophysical Union Fall Meeting , New Orleans, LA.	2021
Primary Convener. <i>NH53 - Emerging Needs and Approaches for Climate Services: Understanding and Developing Innovative Approaches to User-Oriented Climate Services.</i> American Geophysical Union Fall Meeting , San Francisco, CA.	2019
Student Organizer. <i>Earth and Environmental Engineering Student Research Symposium.</i> Columbia University , New York, NY.	2018
Student Organizer. <i>Earth and Environmental Engineering Student Research Symposium.</i> Columbia University , New York, NY.	2017

MEDIA HIGHLIGHTS

"FEMA elabora nuevos mapas de riesgo de inundación ¿Afectará esto tu propiedad?". Telemundo Houston [recorded video interview].	2026
"The Climate-Disaster Scores That Could Make or Break Your Home Sale". The Wall Street Journal — Jean Eaglesham and Nicole Friedman [print].	2026
"Historic Texas Flooding". KEYE-AUS (CBS) [recorded video interview].	2025
"Houston's Morning Show". Fox26 Houston [live video interview].	2025
"Breaking down the force of water in the Texas floods". AP News — Michael Phillis [print].	2025
"Here are some things you can do to be better prepared for major flooding". Associated Press — Caleigh Wells [print].	2025
"ABC13-Luke Jones speaks with James Doss-Gollin, an assistant professor of Civil and Environmental Engineering at Rice University, about the key differences between flooding in Houston and central Texas." ABC13 — Luke Jones [recorded video interview].	2025
"After deadly flooding in Central Texas, state lawmakers look to prevent similar tragedies". KVUE — Daniel Perreault [recorded video interview].	2025
"New Flood Warning System Greenlit Shortly Before Deadly Texas Disaster". The Wall Street Journal — Joseph De Avila [print].	2025
"Questions arise on how emergency warning systems work after Central Texas flood". ABC13 — Tom Abrahams [print].	2025
"Bajo la Amenaza del Golfo (Under the Threat of the Gulf)". Telemundo Houston [recorded video interview].	2025
"Will Texas become too hot for humans?". BBC Future — Sarah Griffiths [print].	2023
"Climate change has sent temperatures soaring in Texas". The Texas Tribune — Erin Douglas, Yuriko Schumacher and Alex Ford [print].	2023
"Texas could have foreseen 2021 cold-wave disaster, new study concludes". Texas Climate News — Bob Henson [print].	2022
"Opinion: The risks of climate change are great - so are the rewards of solving it". Houston Chronicle — Andrew Dessler, James Doss-Gollin, and Katherine Hayhoe [print].	2021
"The False Comfort of Higher Seawalls". The New Republic — Paola Rosa-Aquino [print].	2019
"New Study Shows Promise for Long-Term Weather Forecasts in South America". State of the Planet — Elisabeth Gawthrop [print].	2018

LANGUAGE SKILLS

English: Native

Spanish: Fluent
Portuguese: Proficient

TRAINING AND EXPERIENCE

Strategic Program to Accelerate Researchers in Computing (SPARC) Participant — <i>Natural Hazards Engineering Research Infrastructure (NHERI) DesignSafe-CI.</i>	2025
Panel Fellow — <i>NSF CMMI's Game Changer Academies for Advancing Research Innovation.</i>	2021
Visiting Graduate Researcher — <i>Lamontagne Research Group, Department of Civil and Environmental Engineering, Tufts University, Medford, MA.</i>	2019–2020
Summer School Participant — <i>Fluid Dynamics of Sustainability and the Environment, Cambridge University, Cambridge, England.</i>	2016
Education Policy Intern — <i>Elm City Communities / New Haven Housing Authority, New Haven, CT.</i>	2015
President (2014), Design Lead (2013), Member (2012, 2015) — <i>Engineers Without Borders, Yale Student Chapter, New Haven, CT.</i>	2012–2015
Undergraduate Research Assistant — <i>Lab of Jaehong Kim, Department of Chemical and Environmental Engineering, Yale University, New Haven, CT.</i>	2014–2015
Visiting Undergraduate Researcher — <i>Water and Climate Risk Lab, Department of Hydraulic and Environmental Engineering, Universidade Federal do Ceará, Fortaleza, Brazil.</i>	2014
Mechanical Design Intern — <i>Slingshot Team, DEKA Research \& Development, Manchester, NH.</i>	2013
Undergraduate Research Assistant — <i>Lab of Jan Schroers, Department of Mechanical Engineering and Materials Science, Yale University, New Haven, CT.</i>	2012
Ikatú Agua Intern — <i>Fundación Paraguaya, Asunción, Paraguay.</i>	2012